

API Spec: CrossRef & Journal Matching API

TL;DR

A backend service that resolves PMIDs to journal metadata using the CrossRef REST API, then classifies each journal into a medical specialty through a three-tier cascade against a static ISSN classification dataset (2,322 journals, 35 specialties). These classifications power the market size context and specialty display in the PI Profile and request details. CrossRef requires no API key but mandates a mailto parameter (institutional email). The service is triggered bi-monthly post-enrichment, targeting new PMIDs in the request_publications table, and ensures exhaustive ISSN matching from the semicolon-delimited lists.

Goals

Business Goals

- Classify every PI publication into a medical specialty automatically, requiring no manual journal lookups.
- Deliver actionable market size context for each PI's primary specialty, aiding data teams in strategic market identification.
- Build a comprehensive specialty classification record for all PMIDs in the system to support future ML and data analytics.

User Goals

- Data managers automatically see a PI's primary medical specialty and market data on the PI Profile, with no manual processes.
- Any unclassified journals are clearly flagged for team review, supporting iterative improvement of the classification dataset.

Non-Goals

- Journal prestige scoring is not handled here (covered by the PI Enrichment API via Clarivate JCR and Google Scholar Metrics).
 - This service does not update or expand the ISSN or market size datasets itself—these remain manually curated.
 - Only PMIDs already in the request_publications table (i.e., associated with closed/integrated requests) are classified; open/in-progress requests are out of scope.
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User Stories

- **Backend Service**

- As the backend system, I want to resolve each PMID from request_publications to a DOI and ISSN via CrossRef, so that I have a reliable identifier for specialty matching.
 - As the backend system, I want to match each ISSN to the classification dataset using a tiered approach (exact match, normalized title, fuzzy title), to maximize specialty coverage without increasing false matches.
 - As the backend system, I want to aggregate matched specialties for each PI and mark the plurality as Primary Specialty, so every PI Profile distills to a headline specialty and secondary notes.
 - **Data Manager**
 - As a Data Manager, I want to see a PI's primary specialty and market size (Mode) in their profile, with hover details for Min, Max, Mean, CAGR, Confidence, and Source, to understand their research area's value.
 - **Team Lead**
 - As a Team Lead, I want to see a downloadable list of unclassified journals in the admin panel, so we can expand the classification dataset over time.
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Functional Requirements

- **Trigger & Schedule (Priority: High)**
 - Runs as part of the bi-monthly enrichment cycle:
 - Post-PubMed bi-monthly re-run (after new PMIDs added to request_publications).
 - Post-initial closure run (newly closed requests).
 - Processes:
 - All PMIDs in request_publications without specialty classification.
 - Any PMIDs previously marked Unclassified for possible match updates on dataset changes.
- **CrossRef PMID-to-ISSN Resolution (Priority: High)**
 - Input: PMID from request_publications table.
 - Step 1: If DOI is stored (extracted from PubMed XML), use it—avoids redundant CrossRef calls.
 - Step 2: If DOI available, call GET `https://api.crossref.org/works/{doi}` with `mailto` query parameter.
 - Step 3: If no DOI, fallback to GET `https://api.crossref.org/works?query.bibliographic={journal_name}&filter=issn:{issn_hint}`.
 - Retrieve and store print and electronic ISSNs, journal (container-title), and DOI.
 - Implement 0.5 second delay between requests as courtesy; up to 3 retries on 429/503 with exponential backoff.
 - If CrossRef fails, fallback to PubMed journal name for matching.
- **Three-Tier Journal Matching Cascade (Priority: High)**

1. **Tier 1 — ISSN Exact Match:**
 - Compare all CrossRef ISSNs to every ISSN in the semicolon-delimited ISSN List from the classification dataset (must split on semicolon).
 - On match, record specialty, confidence = EXACT_ISSN, and matched title.
 2. **Tier 2 — Normalized Title Match:**
 - Strip punctuation, leading articles (“The”, “A”, etc.), and common prefixes (“Journal of”, etc.) from both CrossRef and dataset titles.
 - Compare (case-insensitive).
 - On match, record specialty, confidence = NORMALIZED_TITLE, and matched title.
 3. **Tier 3 — Fuzzy Title Match:**
 - Token-based fuzzy match (e.g., RapidFuzz); accept matches with similarity $\geq 85\%$.
 - On match, record specialty, confidence = FUZZY_TITLE, similarity score, matched title.
 4. **Unclassified Fallback:**
 - Record UNCLASSIFIED if all tiers fail; log journal name, CrossRef title, ISSN for review.
- **Specialty Aggregation Per PI (Priority: High)**
 - Aggregate all matched specialties for a PI.
 - Primary Specialty = plurality winner across PMIDs (ties broken by most recent publication).
 - Secondary specialties listed as comma-delimited side note.
 - Exclude Unclassified matches from plurality calculation but track count.
 - Store all in pi_enrichment_cache.
 - **Market Size Context (Priority: High)**
 - Lookup primary specialty in the market sizing dataset (35 specialties).
 - Store Mode, Min, Max, Mean, CAGR, Confidence, Source in pi_enrichment_cache.
 - If not found, store NULLs.
 - If Mode < \$50B AND CAGR > 7%, set niche_journal_bonus_eligible = TRUE (thresholds configurable in admin panel).
 - **ISSN Classification Dataset Handling (Priority: High)**
 - Static dataset loaded at deployment; not refreshed automatically.
 - ISSN List column is semicolon-delimited—must split and index all ISSNs at load time.
 - Create issn_index table for O(1) exact match lookups: one row per ISSN.
 - **Unclassified Journal Logging (Priority: Medium)**
 - Journals not matched after 3 tiers are logged in unclassified_journals table with pmid, PubMed and CrossRef journal names, CrossRef ISSN, date.
 - Admin panel displays this log and download option.

User Experience

Entry Point & First-Time User Experience

- Service runs backend-only in the enrichment cycle—no direct user interaction for setup or onboarding.
- Dataset upload (for ISSN mapping) done via admin panel by Team Lead as needed.

Core Experience

- **Step 1:** New or previously unclassified PMIDs are identified post-enrichment.
 - No user intervention. Trigger logic is automatic.
- **Step 2:** For each PMID, the service checks for a stored DOI; if present, proceeds to CrossRef; else tries fallback logic.
 - If CrossRef succeeds, extracts journal metadata; on failure, falls back as described.
- **Step 3:** Journal is matched across the 3-tier cascade.
 - All ISSNs (after splitting) checked in index (Tier 1); if no match, progress to title matching (Tiers 2 and 3).
 - Specialty, match confidence, and matched title are recorded—or journal is flagged as Unclassified.
- **Step 4:** Per-PI specialty summary is updated.
 - PI Profile displays:
 - Headline specialty with market size Mode;
 - Hover: Min, Max, Mean, CAGR, Confidence, Source;
 - Secondary specialties as a side note;
 - Niche market note if applicable.
- **Step 5:** Admin panel displays any Unclassified journals.
 - Table view, count, download CSV option, and date last updated.

Advanced Features & Edge Cases

- Journals with multiple ISSNs (print/electronic) are each tested against all dataset entries.
- If CrossRef fails, logic falls back safely (with logging for review).
- Fuzzy matching is pre-filtered by first word for performance when needed.
- Manual ISSN dataset updates are immediately used after upload—not deferred until next cycle.

UI/UX Highlights

- Amber flag for unclassified PIs in dashboard;
- Accessible hover tooltips for market stats;
- Admin panel clarity: clear columns, crisp download, last-updated date always visible.
- All error states are explicitly surfaced (never silent).

Narrative

When a new PMID hits the system—via closure, re-enrichment, or a historic backfill—the CrossRef & Journal Matching API gets to work behind the scenes. It fetches rich journal metadata using CrossRef, methodically checks for a specialty match in the ISSN index and journal title list, and, as soon as one is found, links it to a medical specialty. Now, data managers look at a PI's profile and immediately see a headline specialty—perhaps Cardiology or Oncology—backed up by credible market size estimates. If the specialty is niche and high-growth, the tool auto-flags it for business development. When unclassified journals accumulate, they aren't hidden: the admin panel tallies and lists them, so the team can refine the classification set each quarter. Over time, the specialty and market context becomes more accurate and complete, saving hours of manual lookup and letting the team focus their expertise on the few edge cases that matter most.

Success Metrics

- **Classification Rate:** >80% of PMIDs classified (not Unclassified) within 3 months of deployment.
- **CrossRef Resolution Rate:** >90% of PMIDs successfully resolved to ISSN via CrossRef.
- **Tier 1 Match Rate:** >70% of resolved ISSNs matched via exact ISSN (Tier 1).
- **Zero Silent Failures:** 100% of PMIDs have a logged classification status.
- **Admin Dataset Coverage:** No unclassified rate >20% for two consecutive enrichment cycles.
- **Per-PI Metrics:** Every PI with ≥ 3 publications has a non-NULL primary specialty.

User-Centric Metrics

- % of PI Profiles with valid specialty and market context displayed.
- Avg. time from dataset upload to all new publications classified (should be <1 cycle).

Business Metrics

- Reduction in manual journal lookup time by data team.
- Number of new niche/high-growth specialties identified (quarterly cadence).

Technical Metrics

- Up-time/availability during scheduled cycles.
- CrossRef call failure/retry rates.
- Fuzzy match performance (average runtime per batch).

Tracking Plan

- Per PMID: log resolution status, matched tier, and similarity score (if fuzzy), specialty, timestamp.
 - Per PI: store all specialty and market context fields, update date.
 - Per cycle: total processed, tier counts, unclassified count, CrossRef failures.
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Technical Considerations

Technical Needs

- Automated backend job run bi-monthly (with optional manual trigger by admin).
- ISSN classification dataset loader and parser (splits semicolon lists, builds issn_index).
- CrossRef REST API integration with exponential backoff, polite mailto (CROSSREF_MAILTO from env).
- Three-stage matching logic (exact, normalized, fuzzy).
- Database tables: issn_index, journal_classifications, pi_enrichment_cache, unclassified_journals.

Integration Points

- Existing request_publications and PubMed ingestion pipeline (reads DOI/metadata).
- Admin panel for dataset upload, unclassified downloads, and cycle stats.

Data Storage & Privacy

- No sensitive data; stores publication, journal, and specialty metadata only.
- All emails (for the CrossRef polite pool) stored as environment variables, never in code.

Scalability & Performance

- Expected to process batch of new and previously unclassified PMIDs (~100–1,000 per cycle).
- Fuzzy matching pre-filter by journal title's first word to manage cycle runtime.
- Single-index table for ISSN to specialty lookup for O(1) matching.

Potential Challenges

- CrossRef coverage gaps (older or rare journals).
 - Multiple ISSNs and nonstandard journal titles.
 - Dataset staleness; unclassified rate as health check.
 - Fuzzy matching performance at scale.
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Milestones & Sequencing

Project Estimate

- Small-Medium: 4–5 days (1 backend developer).

Team Size & Composition

- 1 backend developer (handles API integration, DB work, admin panel hooks).

Suggested Phases

Phase 1: Dataset Load & CrossRef Integration (Day 1–2)

- Deliverables: Load and index ISSN dataset, implement issn_index, set up CrossRef polite pool access.
- Dependencies: Classification dataset available and tested.

Phase 2: Matching Cascade & Logging (Day 2–3)

- Deliverables: Three-tier matching, unclassified_journals logging, CrossRef fallback handling, exponential backoff.
- Dependencies: issn_index ready.

Phase 3: Specialty Aggregation & Market Context (Day 3–4)

- Deliverables: Aggregate per-PI specialties, implement market size lookup and niche flag, write outputs to appropriate caches/tables.
- Dependencies: Market sizing dataset loaded.

Phase 4: Admin View & QA (Day 4–5)

- Deliverables: Unclassified journal table in admin panel, downloadable CSV, dataset last-updated date, per-cycle stats. Integration into enrichment cycle trigger. Spot QA.
- Dependencies: Previous phases complete.

Developer Callouts

- `CROSSREF_MAILTO` is an environment variable—never hardcoded.
- Always split ISSN lists on semicolon; index each ISSN separately at load.
- Extract and save DOI from PubMed to eliminate unnecessary CrossRef calls.
- Read niche bonus thresholds (\$50B, 7% CAGR) from admin config, never hardcode.
- Watch and alert if unclassified rate >20% post-deployment.
- Fuzzy matching: optimize by pre-filtering on journal title's first word to reduce unnecessary similarity calculations.